# Diagramas de diseño de Nodo de Monitoreo Agrícola de Bajo Consumo

Equipo 7:

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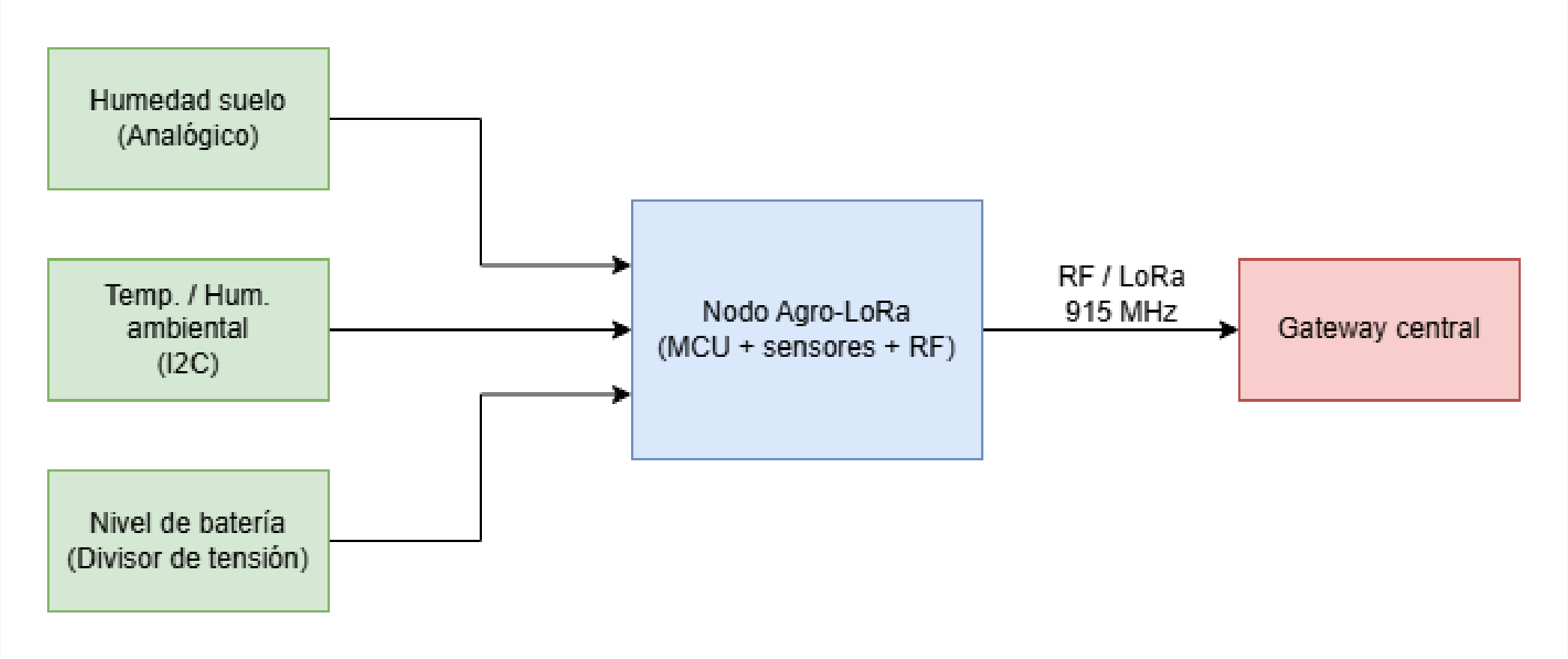
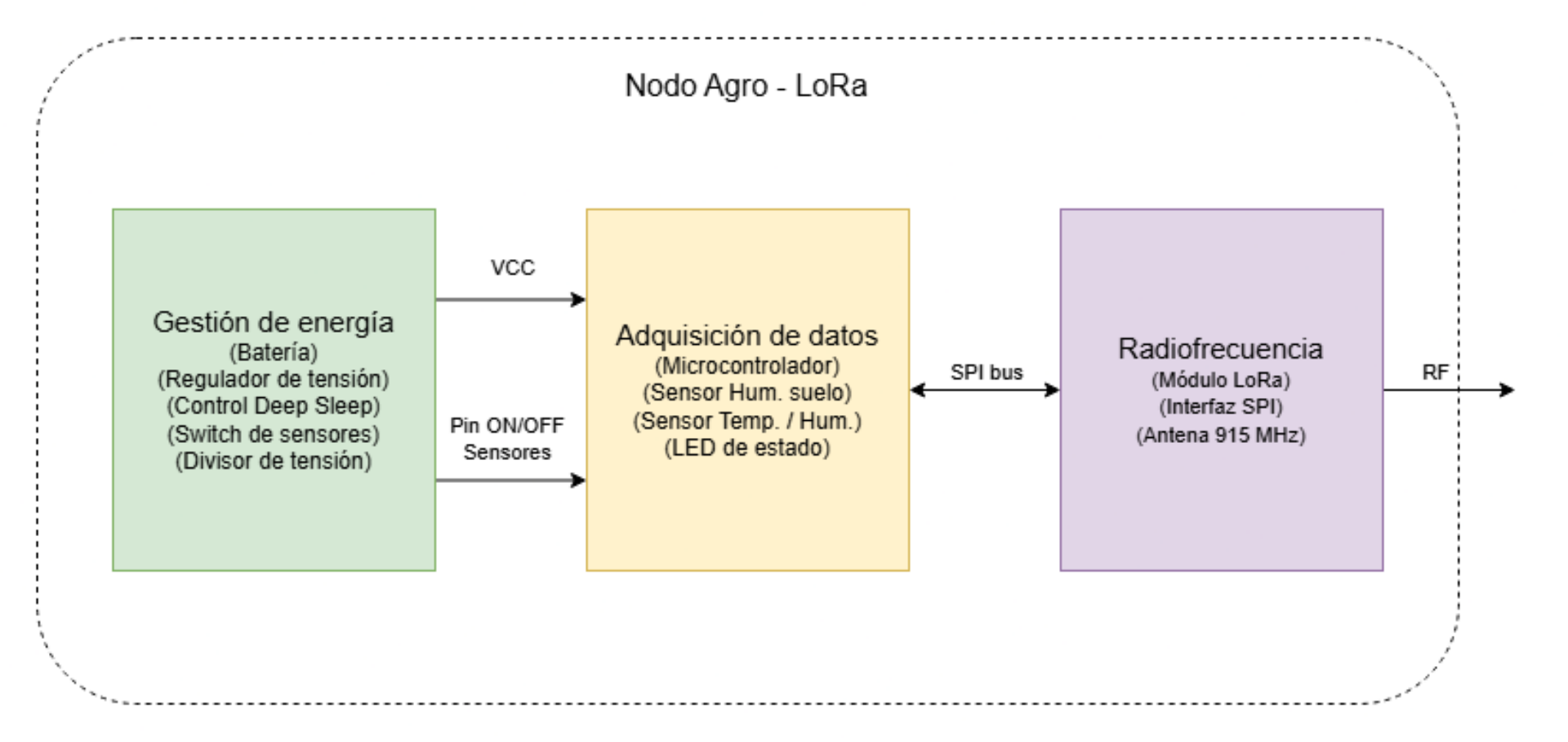
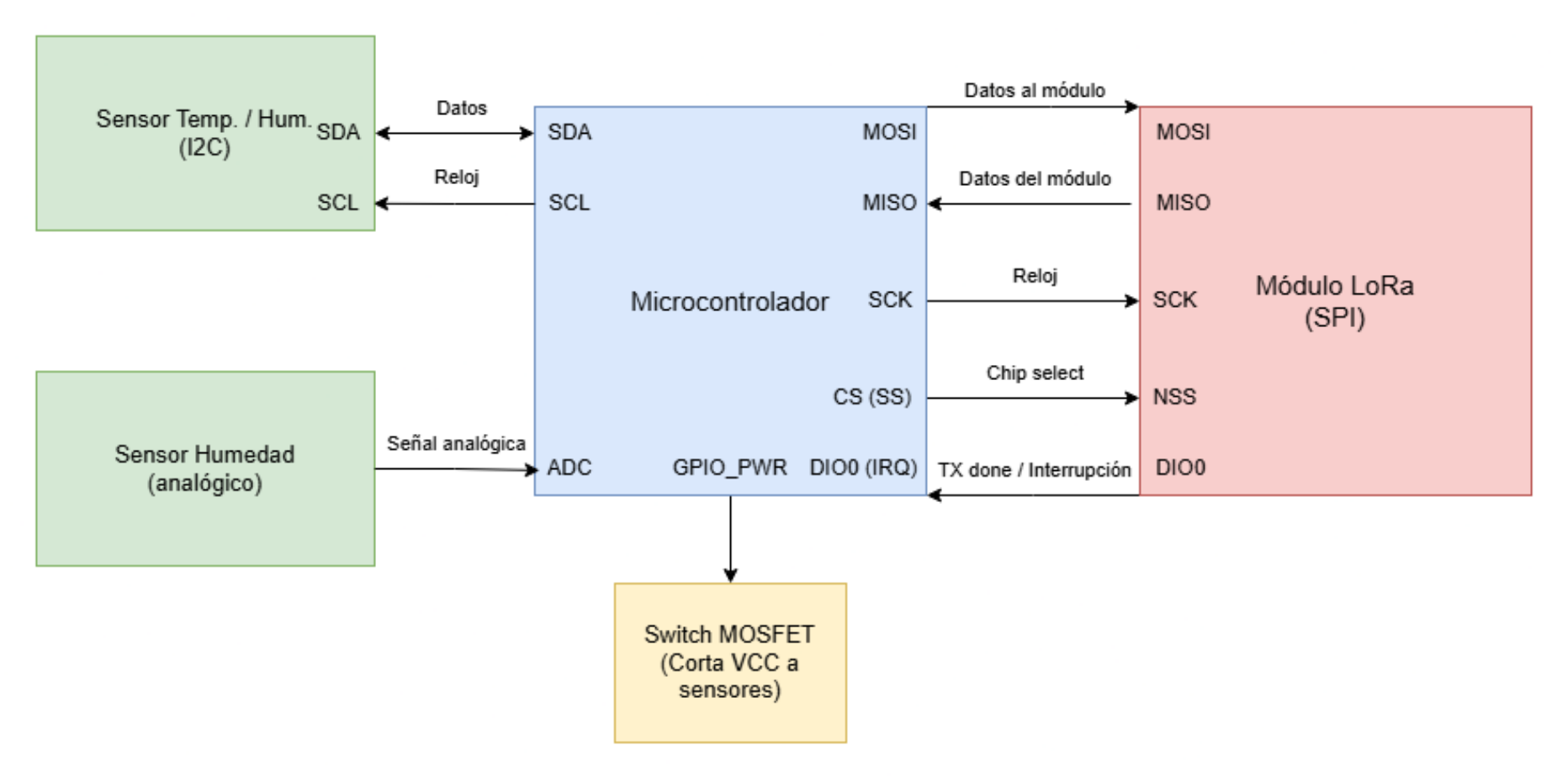
# Diagrama de nivel 1



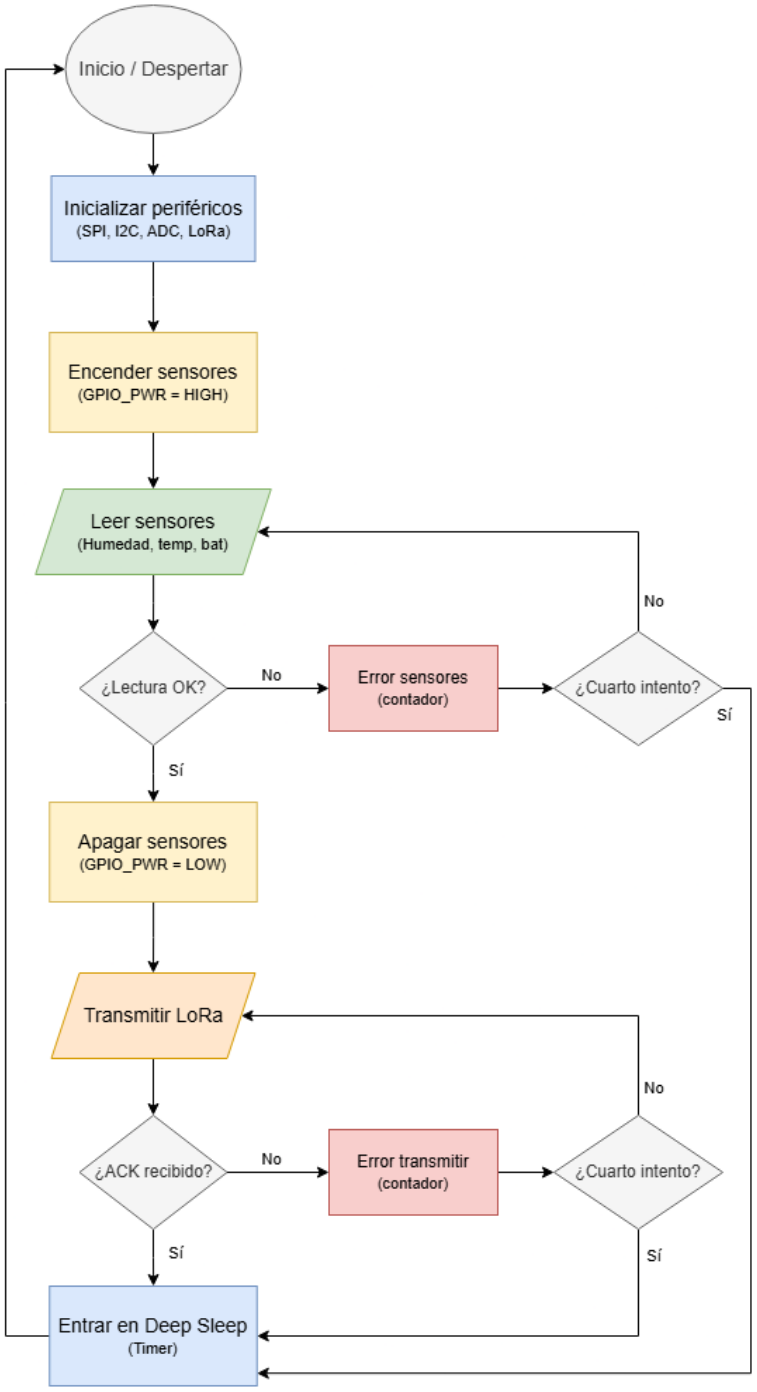
# Diagrama de nivel 2



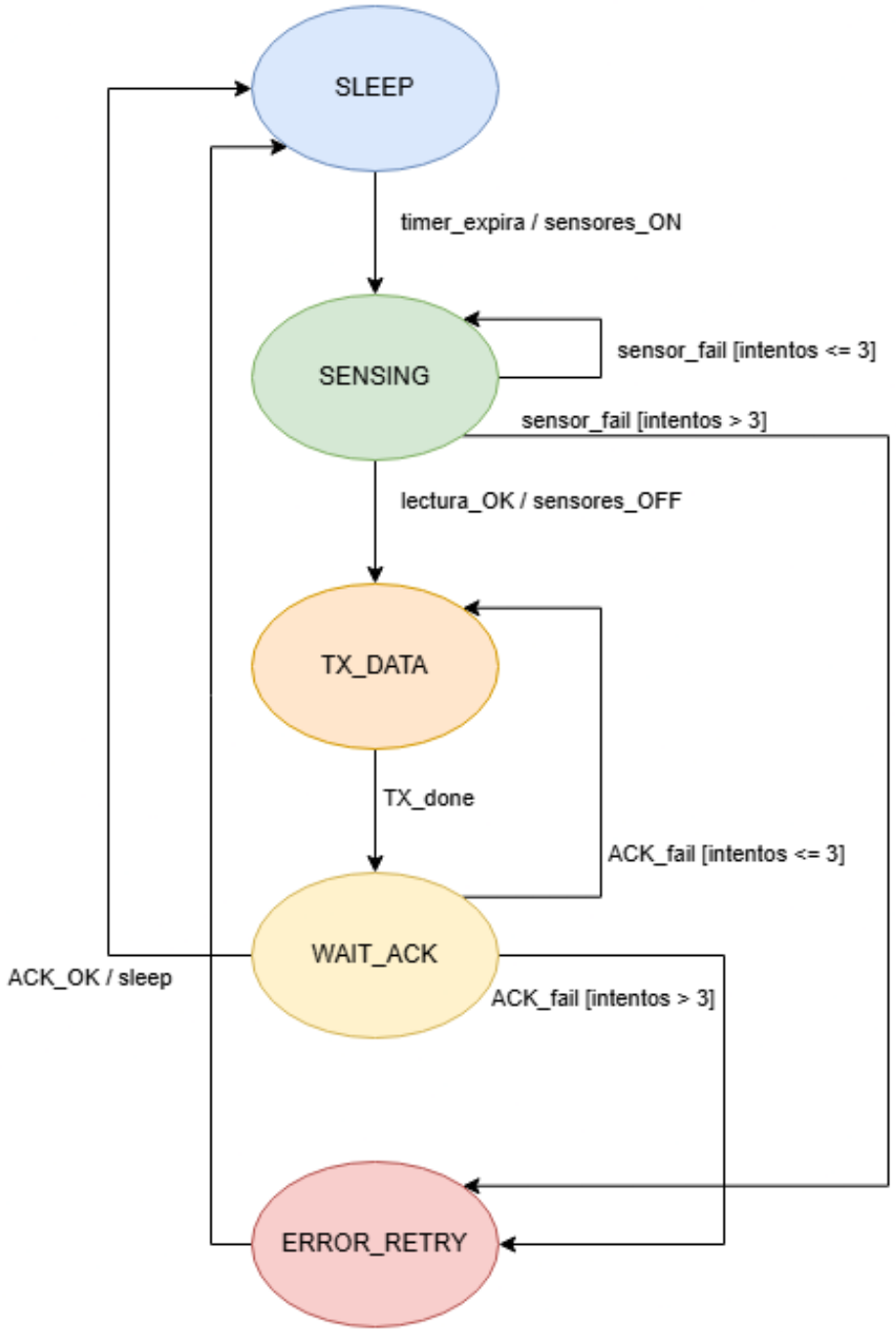
# Diagramas de nivel 3 y 4



- Diagrama de flujo



- Máquina de estados



# Nivel 5 Hardware real

- MCU: ESP32 – Deep Sleep 10 µA, SPI + ADC nativos
- LoRa: RFM95W – 915 MHz, SF12, 125 kHz, 20 dBm
- Sensor T/H: HTU21D (I2C)
- Sensor suelo: capacitivo analógico (ADC)
- Regulador: MCP1700 – 1.6 µA quiescent
- Aislamiento: MOSFET 2N7002

# Caracteristicas consumo

| Modo | Corriente | Tiempo |
|---|---|---|
| Deep Sleep | 12 µA | 95% del ciclo |
| Sensing | 30 mA | ~500 ms |
| TX LoRa | 120 mA | ~1-2 s |

**Vida útil estimada con batería 2000 mAh: +6 meses**